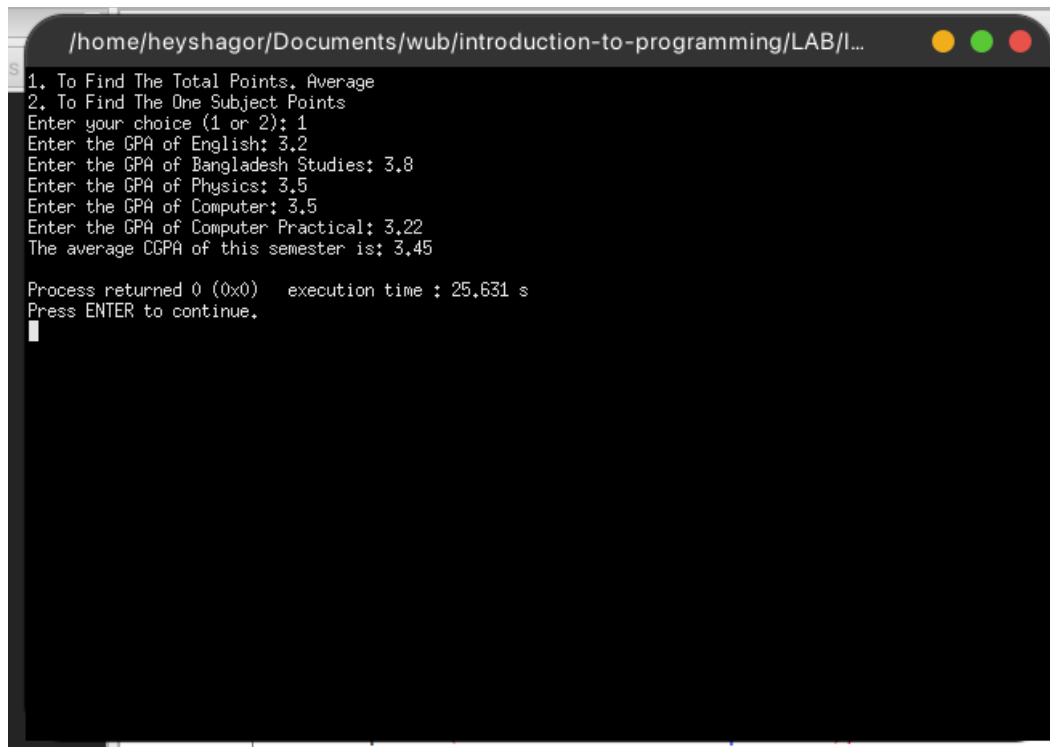
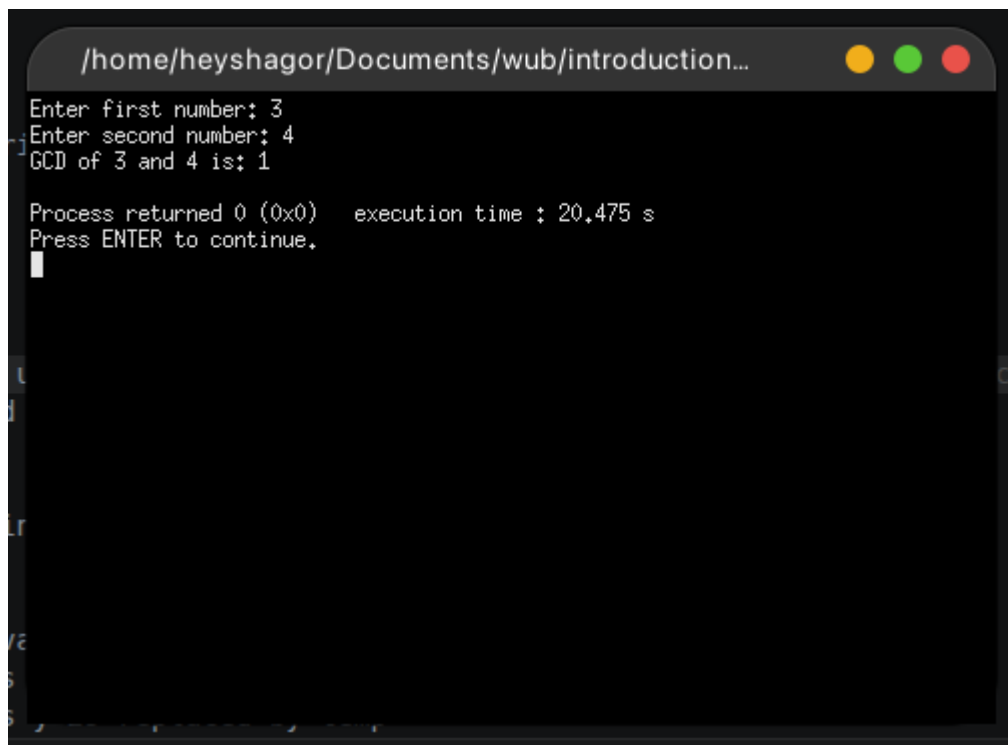


Lab Code 01 Output:A terminal window with a dark background and light gray text. The window title bar shows the path "/home/heyshagor/Documents/wub/introduction-to-programming/LAB/l...". The output text is as follows:

```
1. To Find The Total Points, Average
2. To Find The One Subject Points
Enter your choice (1 or 2): 1
Enter the GPA of English: 3.2
Enter the GPA of Bangladesh Studies: 3.8
Enter the GPA of Physics: 3.5
Enter the GPA of Computer: 3.5
Enter the GPA of Computer Practical: 3.22
The average CGPA of this semester is: 3.45

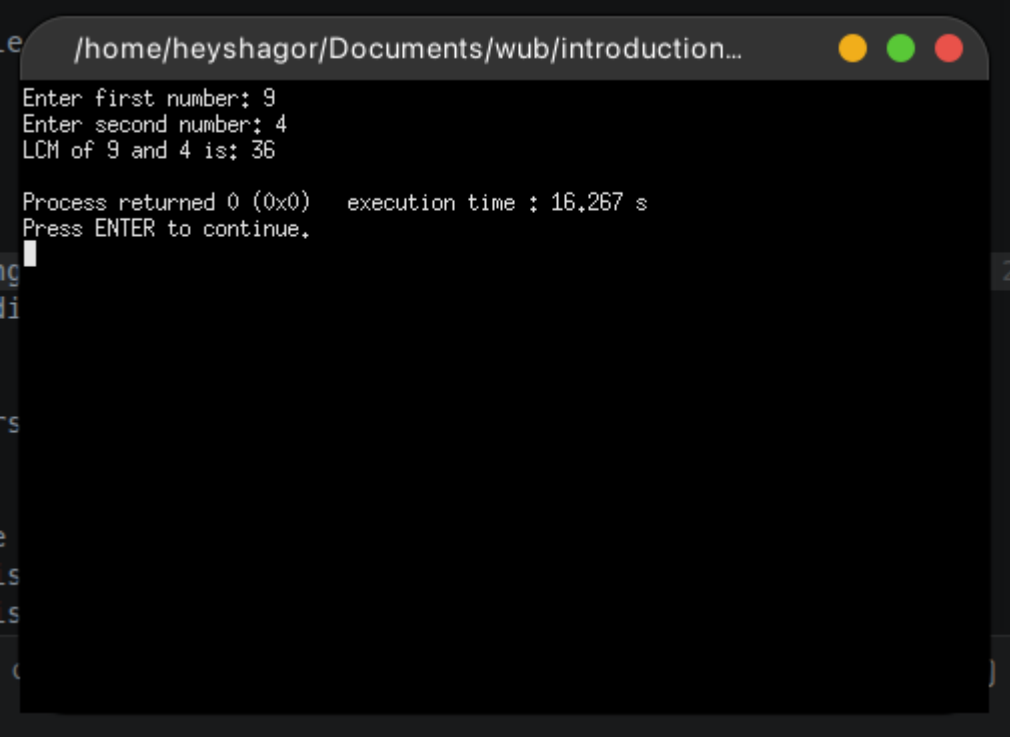
Process returned 0 (0x0)   execution time : 25.631 s
Press ENTER to continue.
```


Lab Code 02 Output:A terminal window with a dark background and light gray text. The window title bar shows the path "/home/heyshagor/Documents/wub/introduction...". The output text is as follows:

```
Enter first number: 3
Enter second number: 4
GCD of 3 and 4 is: 1

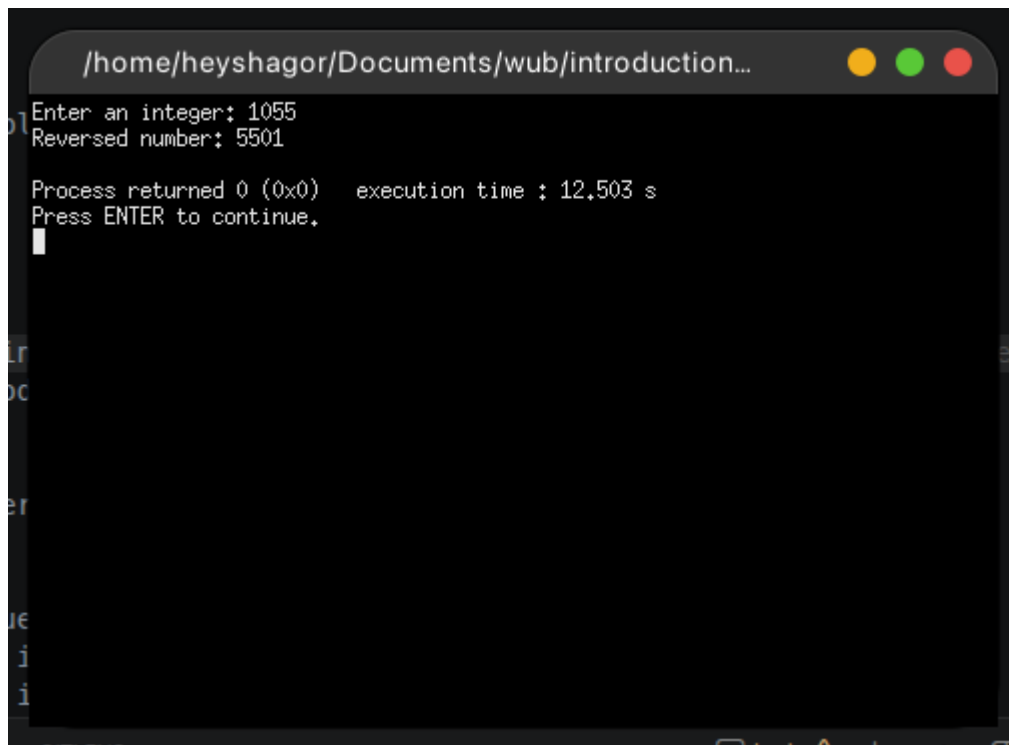
Process returned 0 (0x0)   execution time : 20.475 s
Press ENTER to continue.

```


Lab Code 03 Output:A screenshot of a terminal window with a dark background. The window title bar shows the path "/home/heyshagor/Documents/wub/introduction...". The terminal displays the following text: "Enter first number: 9", "Enter second number: 4", "LCM of 9 and 4 is: 36", "Process returned 0 (0x0) execution time : 16.267 s", and "Press ENTER to continue." followed by a cursor on a new line.

```
/home/heyshagor/Documents/wub/introduction...
Enter first number: 9
Enter second number: 4
LCM of 9 and 4 is: 36

Process returned 0 (0x0) execution time : 16.267 s
Press ENTER to continue.
█
```


Lab Code 04 Output:

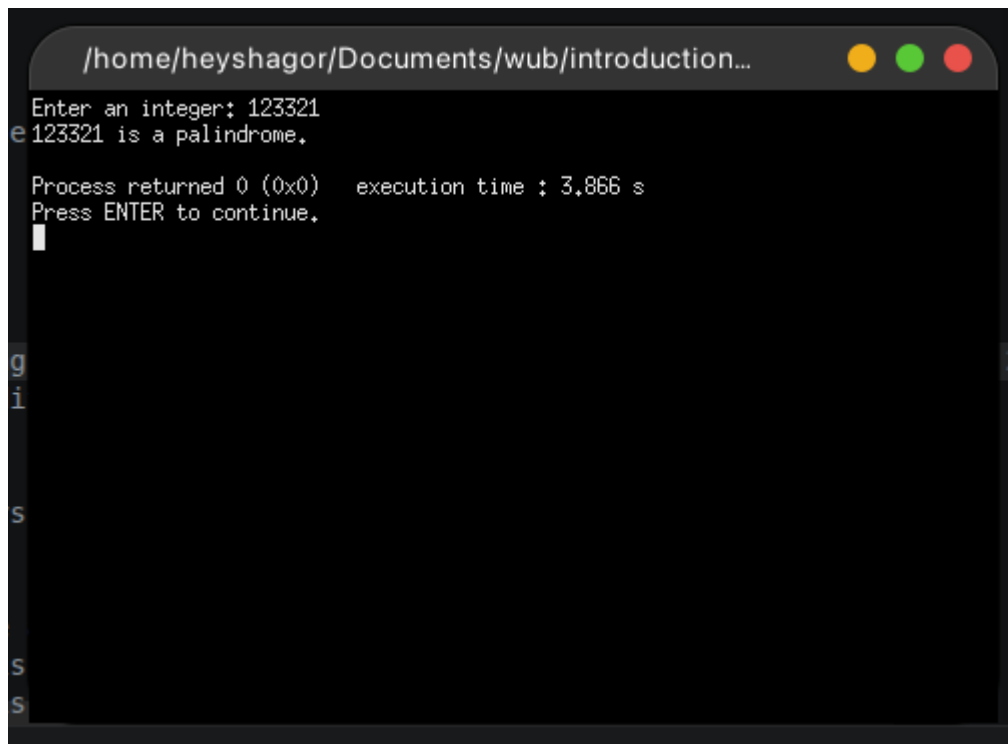
A terminal window with a dark background and a title bar showing the path `/home/heyshagor/Documents/wub/introduction...`. The window contains the following text:

```
Enter an integer: 1055
Reversed number: 5501

Process returned 0 (0x0)   execution time : 12.503 s
Press ENTER to continue.

```

A cursor is visible on the line following "Press ENTER to continue.".

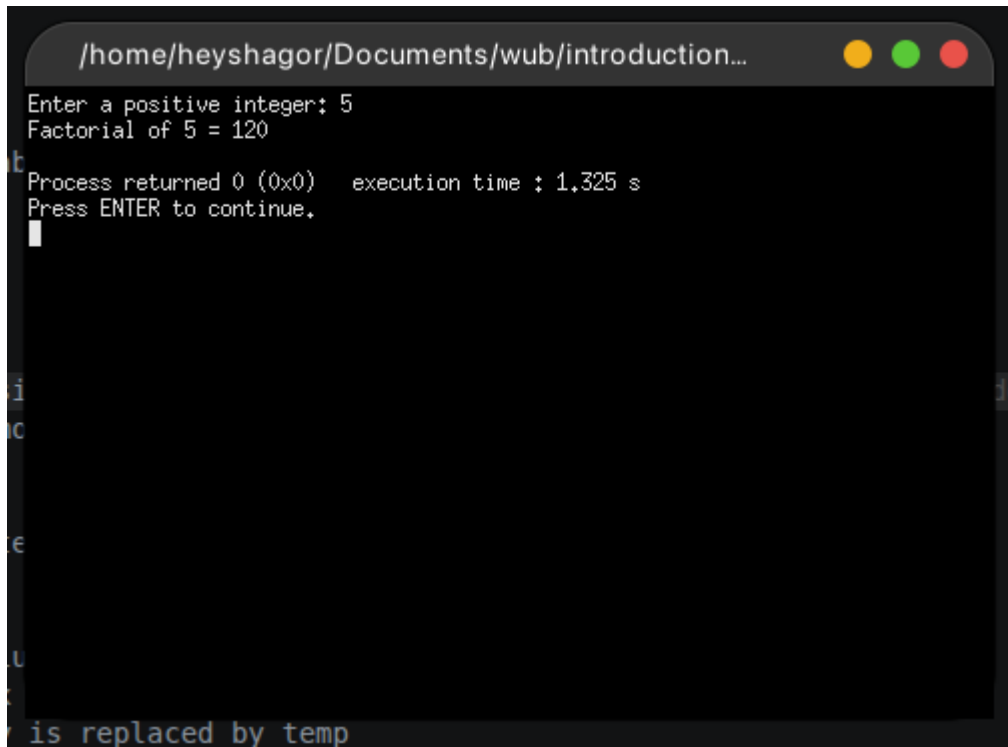
Lab Code 05 Output:

A terminal window with a dark background and a title bar showing the path `/home/heyshagor/Documents/wub/introduction...`. The window contains the following text:

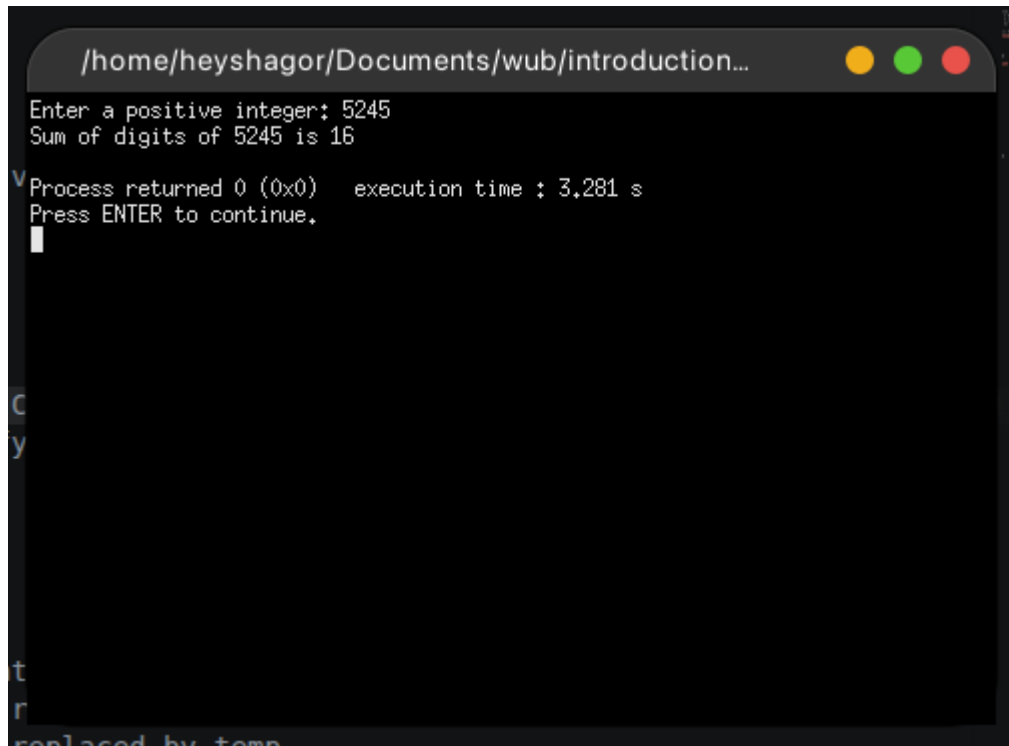
```
Enter an integer: 123321
123321 is a palindrome.

Process returned 0 (0x0)   execution time : 3.866 s
Press ENTER to continue.
█
```

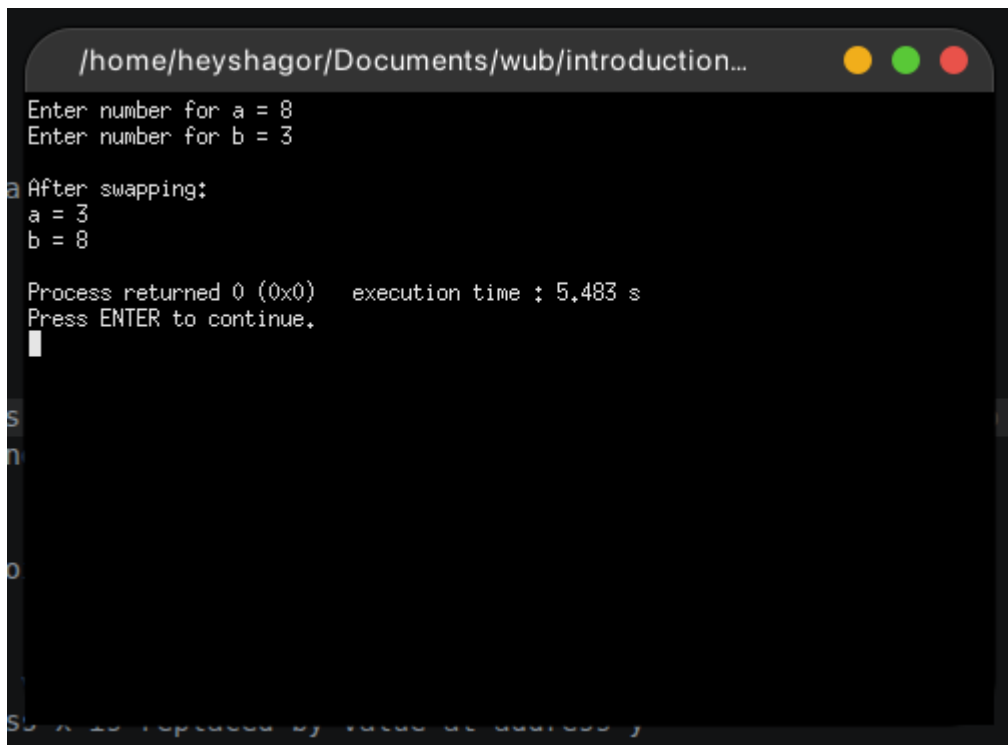
On the left side of the terminal window, there is a vertical scrollbar with labels 'g', 'i', 's', 's', 's' and a page number '2'.

Lab Code 06 Output:

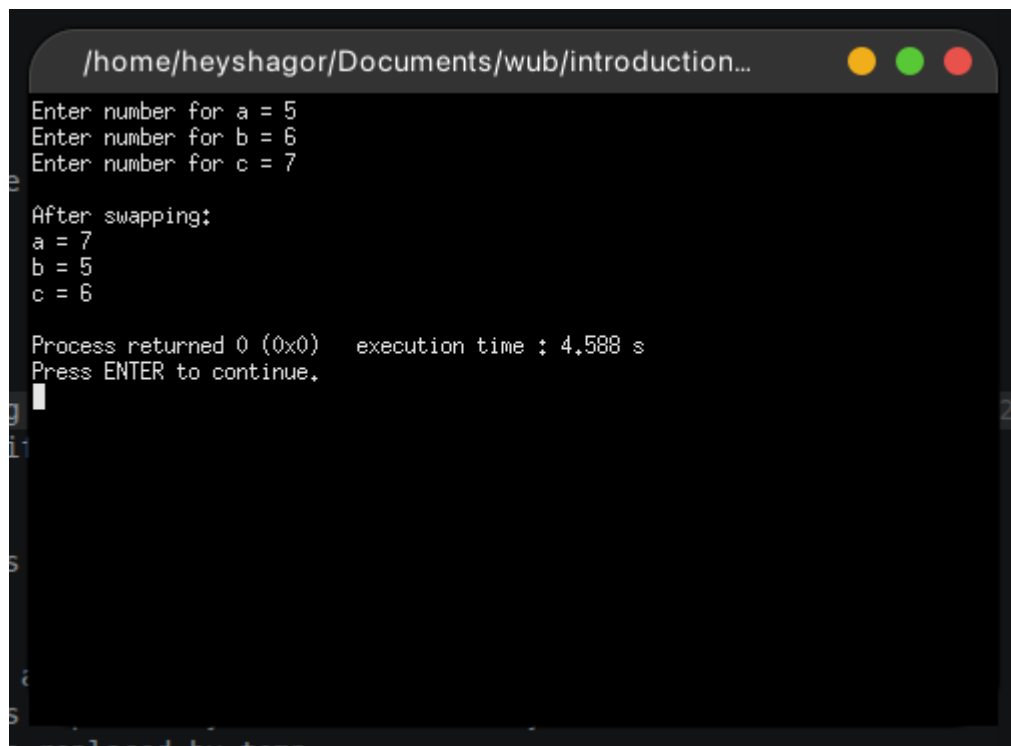
```
/home/heyshagor/Documents/wub/introduction...
Enter a positive integer: 5
Factorial of 5 = 120
Process returned 0 (0x0)   execution time : 1.325 s
Press ENTER to continue.
█
is replaced by temp
```


Lab Code 07 Output:A screenshot of a terminal window with a dark background. The window title bar shows the path "/home/heyshagor/Documents/wub/introduction...". The terminal displays the following text: "Enter a positive integer: 5245", "Sum of digits of 5245 is 16", "v Process returned 0 (0x0) execution time : 3.281 s", and "Press ENTER to continue.". A white cursor is visible on the line "Press ENTER to continue.". On the left side of the terminal, there are some faint, partially visible characters: "c", "y", "t", "r", and "replaced by temp".

```
/home/heyshagor/Documents/wub/introduction...
Enter a positive integer: 5245
Sum of digits of 5245 is 16
v Process returned 0 (0x0) execution time : 3.281 s
Press ENTER to continue.
█
```


Lab Code 08 Output:

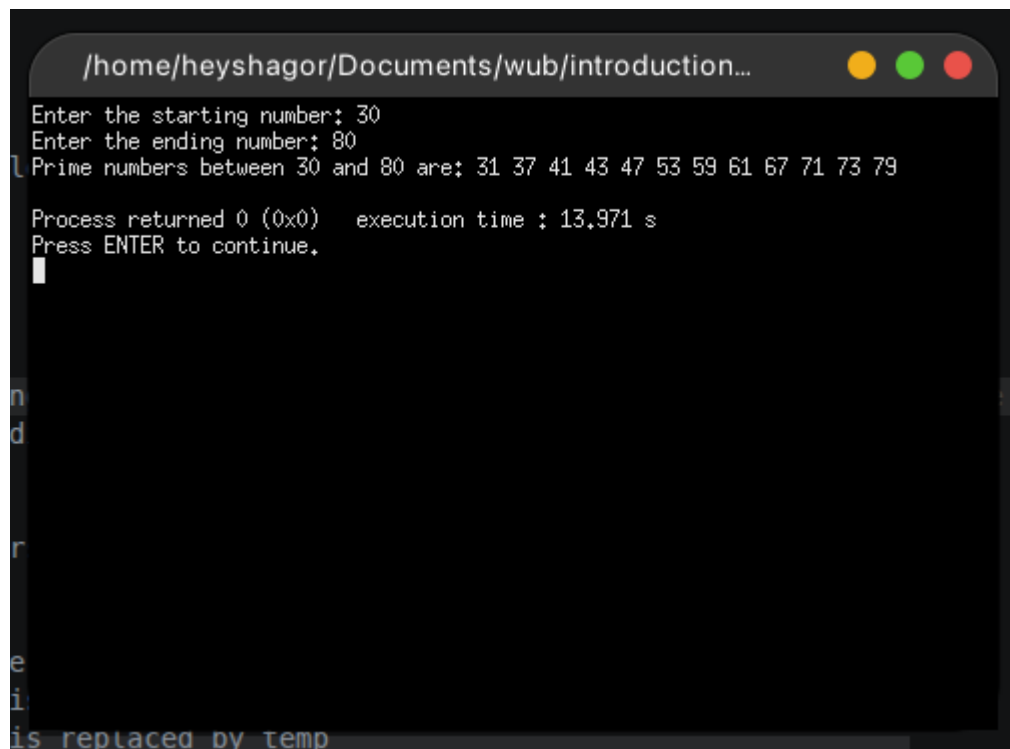
```
/home/heyshagor/Documents/wub/introduction...
Enter number for a = 8
Enter number for b = 3
After swapping:
a = 3
b = 8
Process returned 0 (0x0)   execution time : 5.483 s
Press ENTER to continue.
█
```


Lab Code 09 Output:

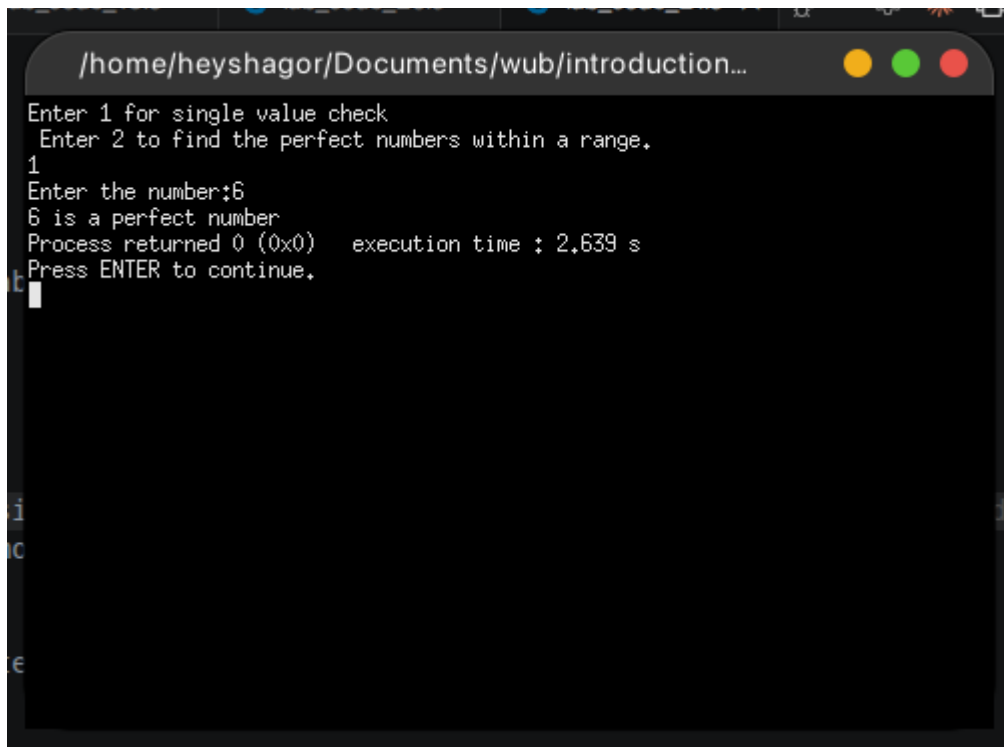
```
/home/heyshagor/Documents/wub/introduction...
Enter number for a = 5
Enter number for b = 6
Enter number for c = 7

After swapping:
a = 7
b = 5
c = 6

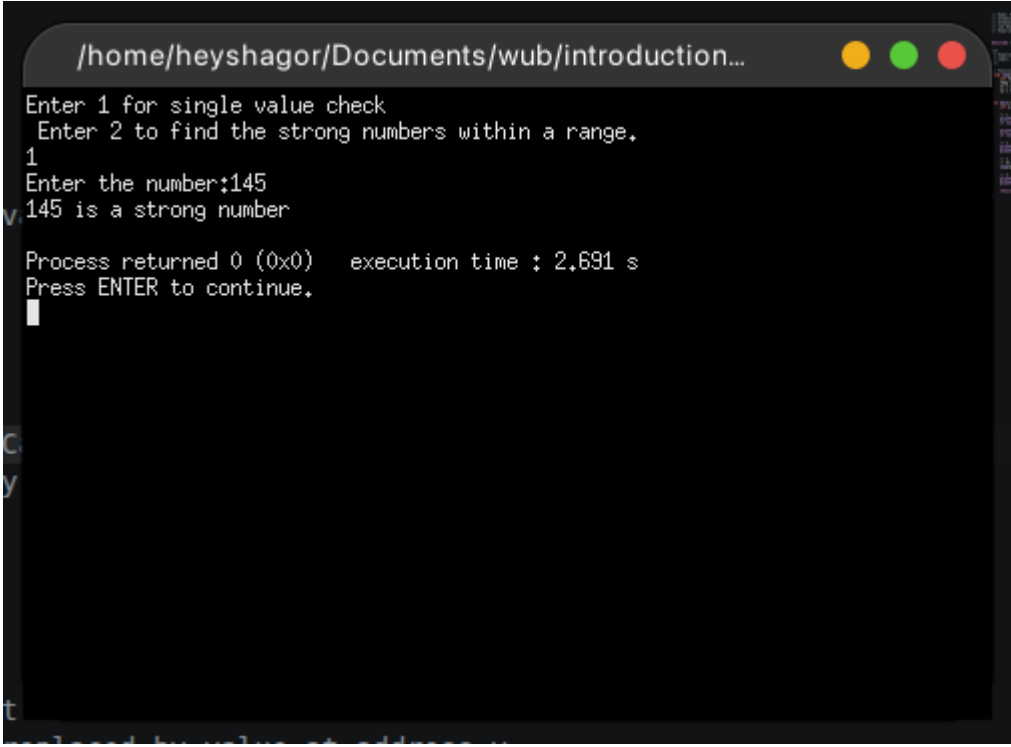
Process returned 0 (0x0)   execution time : 4.588 s
Press ENTER to continue.
```


Lab Code 10 Output:

```
/home/heyshagor/Documents/wub/introduction...  
Enter the starting number: 30  
Enter the ending number: 80  
Prime numbers between 30 and 80 are: 31 37 41 43 47 53 59 61 67 71 73 79  
  
Process returned 0 (0x0)   execution time : 13.971 s  
Press ENTER to continue.  
█  
  
n  
d  
  
r  
  
e  
i  
  
is replaced by temp
```

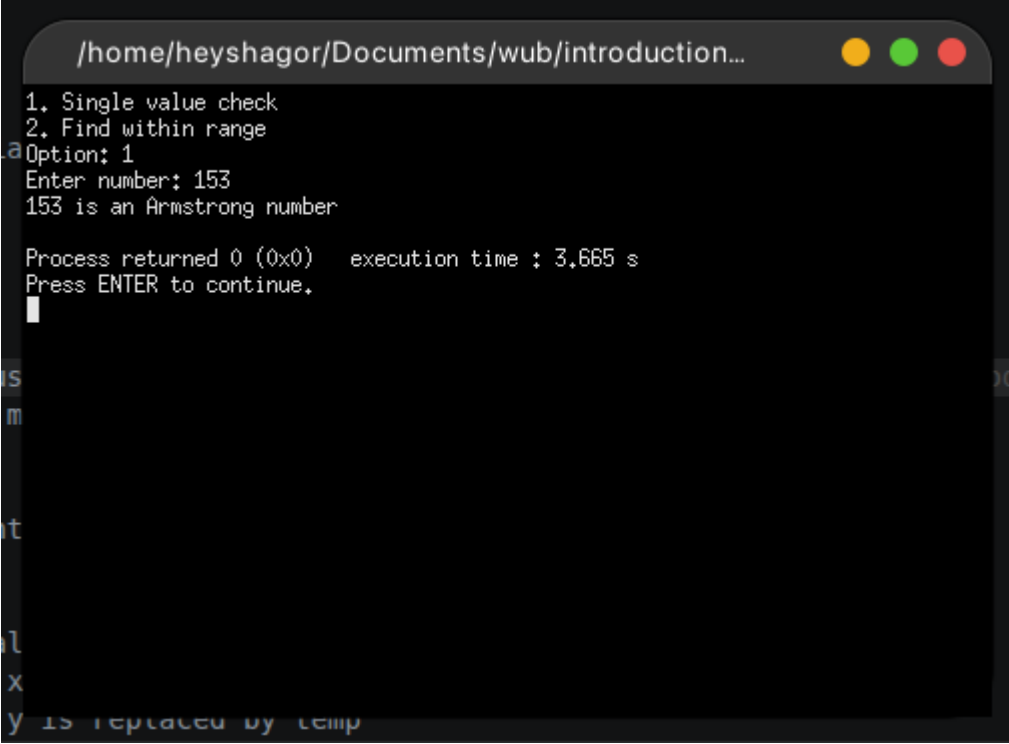

Lab Code 11 Output:

```
/home/heyshagor/Documents/wub/introduction...
Enter 1 for single value check
Enter 2 to find the perfect numbers within a range.
1
Enter the number:6
6 is a perfect number
Process returned 0 (0x0) execution time : 2.639 s
Press ENTER to continue.
█
```


Lab Code 12 Output:A screenshot of a terminal window with a dark background. The window title bar shows the path "/home/heyshagor/Documents/wub/introduction..." and three window control buttons (yellow, green, red). The terminal text is as follows:

```
Enter 1 for single value check
Enter 2 to find the strong numbers within a range.
1
Enter the number:145
145 is a strong number

Process returned 0 (0x0)   execution time : 2.691 s
Press ENTER to continue.
█
```


Lab Code 13 Output:

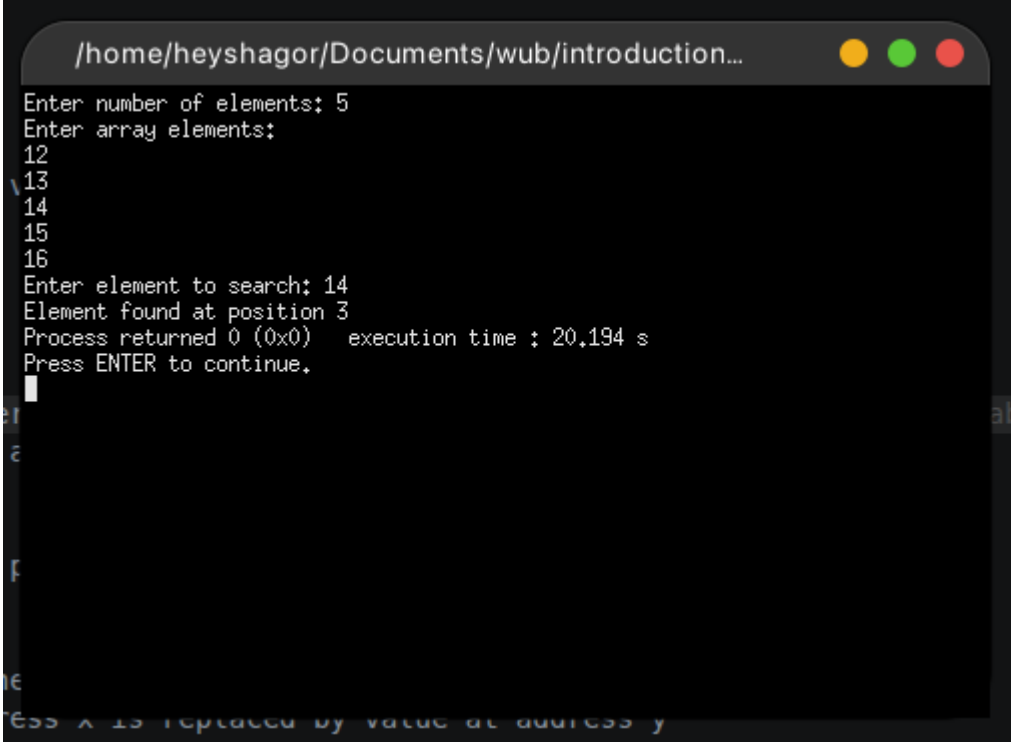
```
/home/heyshagor/Documents/wub/introduction...
1. Single value check
2. Find within range
a Option: 1
Enter number: 153
153 is an Armstrong number

Process returned 0 (0x0)   execution time : 3.665 s
Press ENTER to continue.
█

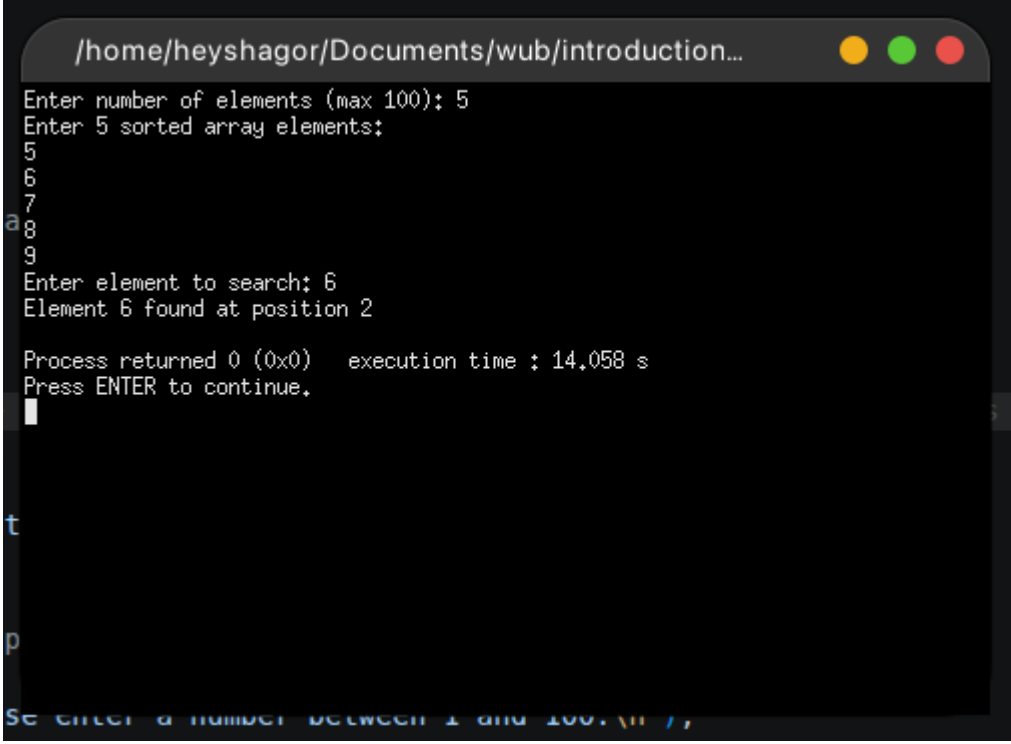
S
m

t

l
x
y is replaced by temp
```


Lab Code 14 Output:

```
/home/heyshagor/Documents/wub/introduction...
Enter number of elements: 5
Enter array elements:
12
13
14
15
16
Enter element to search: 14
Element found at position 3
Process returned 0 (0x0)   execution time : 20.194 s
Press ENTER to continue.
█
```


Lab Code 15 Output:

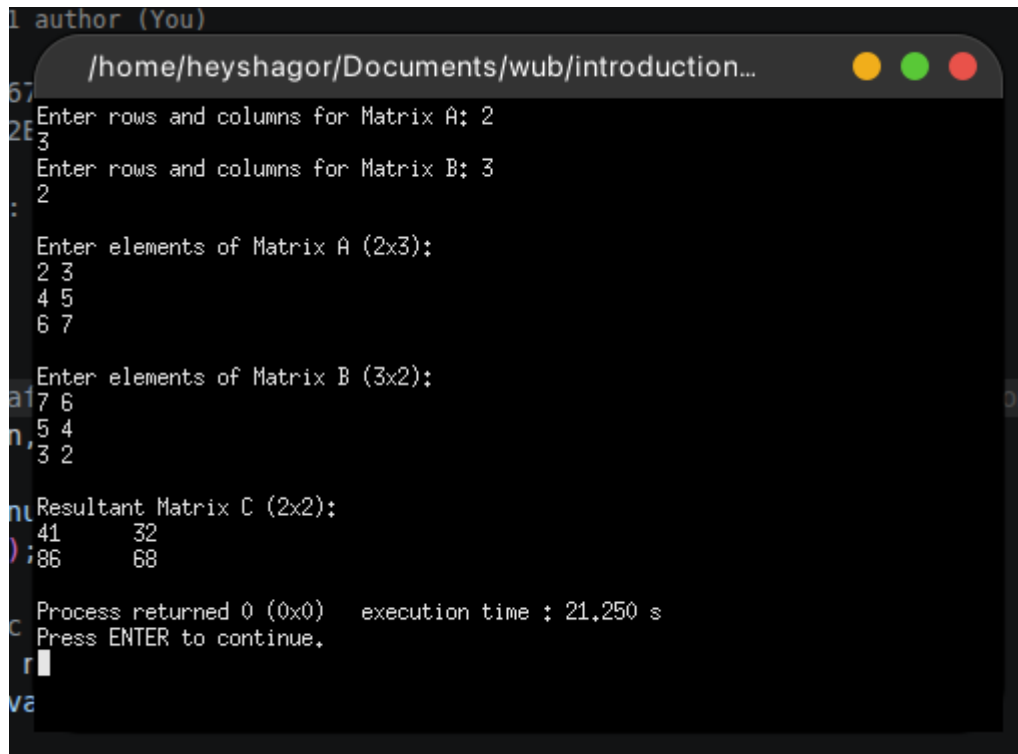
```
/home/heyshagor/Documents/wub/introduction...
Enter number of elements (max 100): 5
Enter 5 sorted array elements:
5
6
7
8
9
Enter element to search: 6
Element 6 found at position 2

Process returned 0 (0x0)   execution time : 14.058 s
Press ENTER to continue.
t
p
Se enter a number between 1 and 100. \n //,
```

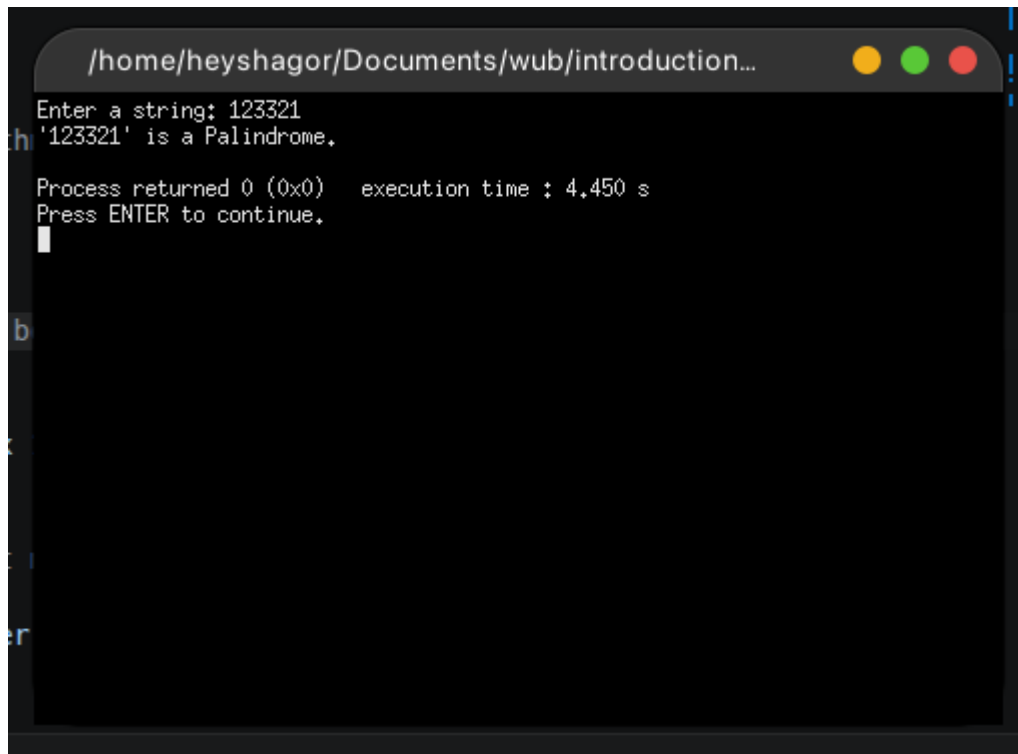

Lab Code 16 Output:

```
/home/heyshagor/Documents/wub/introduction...
Enter rows and columns (max 10x10): 4
5
Enter matrix elements (4 x 5):
Element [0][0]: 4 5 3 4
Element [0][1]: Element [0][2]: Element [0][3]: Element [0][4]: 5 8 9 7
Element [1][0]: Element [1][1]: Element [1][2]: Element [1][3]: 1 8 6 7
Element [1][4]: Element [2][0]: Element [2][1]: Element [2][2]: 1 5 8 4
Element [2][3]: Element [2][4]: Element [3][0]: Element [3][1]: 1 3 5 4
Element [3][2]: Element [3][3]: Element [3][4]:
Original Matrix:
4      5      3      4      5
8      9      7      1      8
6      7      1      5      8
4      1      3      5      4
Transposed Matrix:
4      8      6      4
5      9      7      1
3      7      1      3
4      1      5      5
5      8      8      4
! Process returned 0 (0x0)   execution time : 40.842 s
Press ENTER to continue.

```

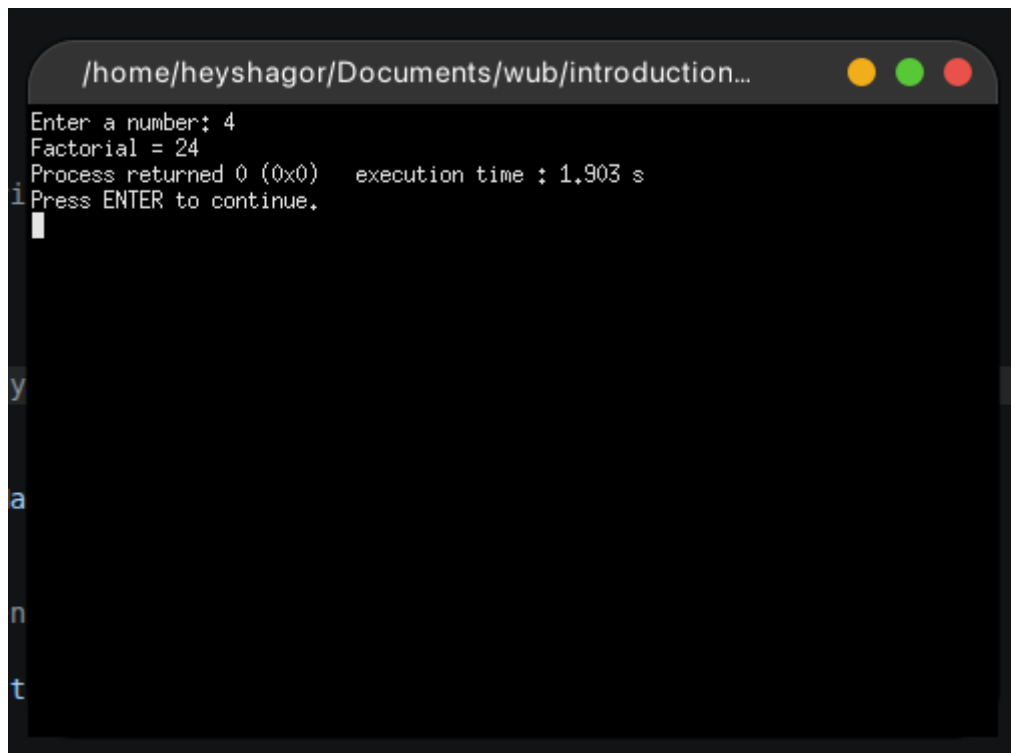

Lab Code 17 Output:

```
1 author (You)
/home/heyshagor/Documents/wub/introduction...
67
2 Enter rows and columns for Matrix A: 2
3
4 Enter rows and columns for Matrix B: 3
5
6 :
7
8 Enter elements of Matrix A (2x3):
9 2 3
10 4 5
11 6 7
12
13 Enter elements of Matrix B (3x2):
14 7 6
15 5 4
16 3 2
17
18 Resultant Matrix C (2x2):
19 41 32
20 86 68
21
22 Process returned 0 (0x0)   execution time : 21.250 s
23 Press ENTER to continue.
24
25
```


Lab Code 18 Output:

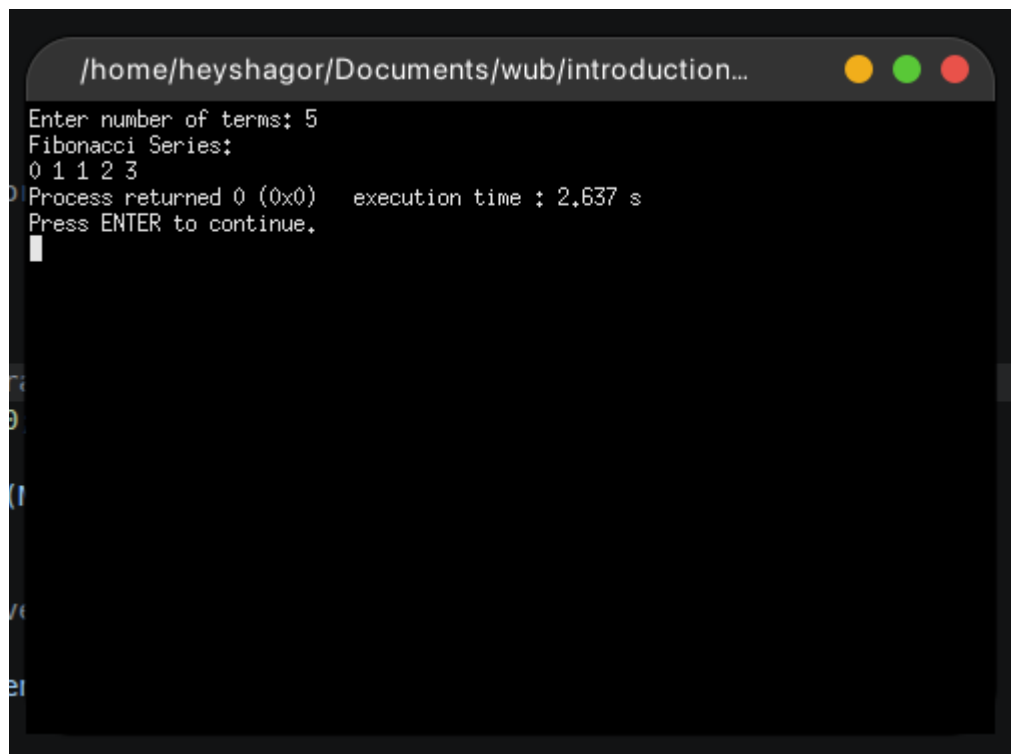
A terminal window with a dark background and a title bar. The title bar contains the path `/home/heyshagor/Documents/wub/introduction...` and three window control buttons (yellow, green, red). The terminal displays the following text:

```
Enter a string: 123321
'123321' is a Palindrome.
Process returned 0 (0x0)   execution time : 4.450 s
Press ENTER to continue.
█
```

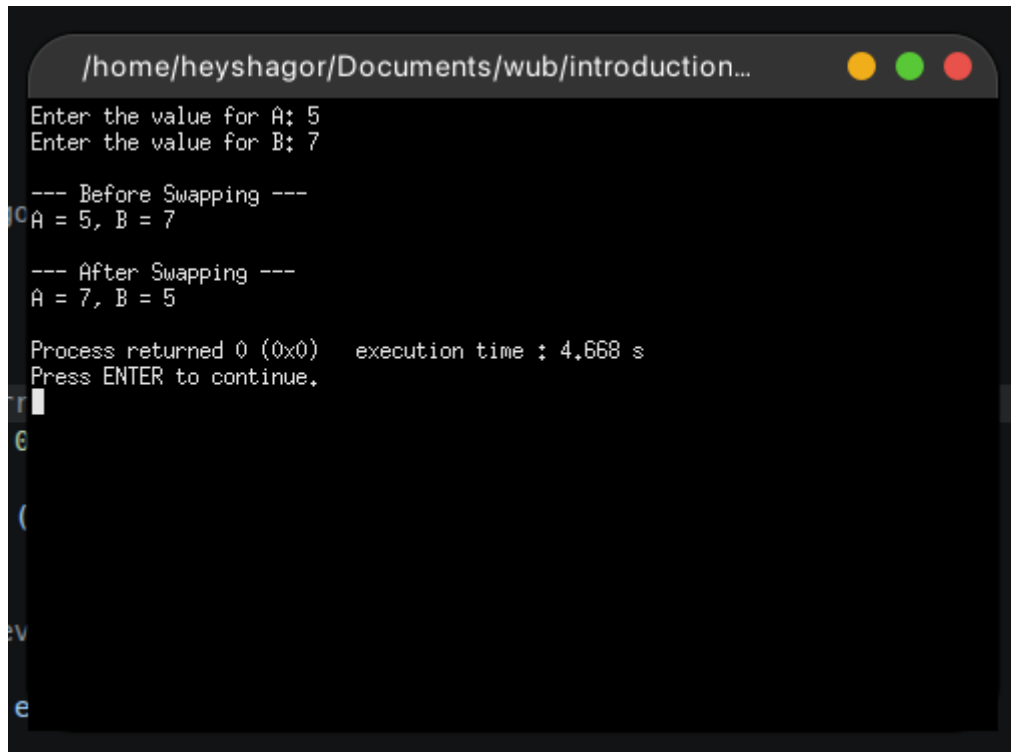

Lab Code 19 Output:A terminal window with a dark background and light gray text. The window title bar shows the path "/home/heyshagor/Documents/wub/introduction...". The output text is as follows:

```
Enter a number: 4
Factorial = 24
Process returned 0 (0x0)   execution time : 1.903 s
Press ENTER to continue.
```

A white cursor is visible on the line "Press ENTER to continue."

Lab Code 20 Output:A terminal window with a dark background and light gray text. The window title bar shows the path "/home/heyshagor/Documents/wub/introduction...". The terminal output displays the Fibonacci series for 5 terms. The text is as follows:

```
/home/heyshagor/Documents/wub/introduction...
Enter number of terms: 5
Fibonacci Series:
0 1 1 2 3
Process returned 0 (0x0)   execution time : 2.637 s
Press ENTER to continue.
█
```


Lab Code 21 Output:

```
/home/heyshagor/Documents/wub/introduction...
Enter the value for A: 5
Enter the value for B: 7

--- Before Swapping ---
A = 5, B = 7

--- After Swapping ---
A = 7, B = 5

Process returned 0 (0x0)   execution time : 4.668 s
Press ENTER to continue.
```